Nicolas Erbetti

- Robotics Engineer position -

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In a nutshell

- France, Japan & Luxembourg
- Worked in international and multilingual teams for ~4yrs.

Computer Science

C/C++
★★★★☆
Python
★★★☆☆
Javascript
★★☆☆☆
C#
★☆☆☆☆
Java
★☆☆☆☆

Languages

French
★★★★★
Native language

English
★★★★☆
Professional level
TOEFL score: 109/120
TOEIC score: 985/990

Japanese
★★☆☆☆
Intermediate level
Equivalent JLPT N4

German
★☆☆☆☆
Certified level: European B1

Professional experiences

October 2023
up until now

Lead Autonomous Stack, Software Architect, Vision & Control
Solarcleano | Grass, Luxembourg

July 2024
up until now

Lead Autonomous Stack Engineer for TALOS, a Horizon Europe funded project
Solarcleano / Energias de Portugal | Grass, Luxembourg / Alqueva, Portugal
- Leading a team of 3 person for the automation part of the work package 3.4 of the TALOS project, dedicated to cleaning a floating photovoltaic plant in Portugal.
- Working with EDP and other members of the European Consortium, such as Alisys.
- Wrote specifications and designed C++ code, state machine, and dual ROS1/ROS2 architecture; implemented kinematic models, UDP PLC communication, FRAMOS depth-cameras with YOLO+Apriltag perception pipeline, extended Kalman filter for localization, PID-based navigation, ROS websocket-based UI for video feedback and remote control, and an MQTT-enabled planification node.
- Training a reinforcement learning policy with a sim2real approach for row switch.

April 2025
up until now

Project coordinator for the automation of a differential-drive robot
Solarcleano / University of Luxembourg | Grass / Luxembourg City, Luxembourg
- Communicating with the team at the University of Luxembourg in charge of the software of a Jetson Orin-powered AGV following autonomously another robot.
- Sharing our codebase for differential-drive robots to jumpstart the development.
- Supplying them when they need specific hardware (plc, camera, lidars...)

October 2023
to January 2025

Computer vision-based control of a 1-ton autonomous robot
Solarcleano | Grass, Luxembourg
- Established a solar panel segmentation pipeline in ROS1 and C++ with the PCL library and its RANSAC algorithm to process Lidars data. Filtered the raw results with Kalman filters, MonteCarlo estimations and geometrical constraints.
- Developed state machine architectures and PID controllers with ROS actions.
- Used the yaw of the segmented panel and a PID to write a realignment node for the navigation stack. Used the panel tilt with PIDs to regulate the robot's brush.
- Created a mapping tool for photovoltaic plants with JavaScript and Maps API.

October 2023
to March 2024

Process improvements
Solarcleano | Grass, Luxembourg
- Set up a docker image to develop ROS melodic code on Ubuntu 22.04 laptops.
- Introduced rostest with our ROS packages to write C++ unit tests with GTest.
- Set up a CI/CD pipeline with github actions to automatically run said tests.

January 2023 to
August 2023

Reinforcement learning framework for the control of a robotic arm
Omron Sinic X | Tokyo, Japan
- Contributed to a dual simulator framework and used it to teach a 6-dof Universal Robot 5 to autonomously cut through soft food with RL methods over a ROS network.
- Calibrated this system with optimization tools (ADAM, Optuna...) to reproduce realistic force/torque data profile obtained with real-world sensor.
- Developed a stable diffusion model to augment the training dataset of the Q-Policy.
- Co-authored a paper on the topic which won the prize for best poster at ICRA 2024.

April 2022 to
August 2022

Data fusion for the control of autonomous underwater & surface robots
Notilo Plus | Marseille, France
- Developed automatic yaw, altitude, and position PID controllers in Python/C++ on ROS for the missions of an AUV; using GPS, pressure, and bathymetric sensors.
- Conducted the unit and integration tests first in the Gazebo simulation and workbench before validating them in real conditions at Marseille's harbor.
- Designed a state-space controller to remote-control an holonomic ASV.

March 2021 to
August 2021

Integration of an ultrasonic sensors belt to a mobile robot
Enova Robotics | Ivry-sur-Seine, France
- Conceived a sonar system as well as a safer mobility solution within ROS for the PGUARD robot. I also designed the mechanical/electrical parts of the prototype.
- Integrated the Ouster OS0 Lidar both in the Gazebo simulation and in reality.

Tools

- ROS / ROS2 / Gazebo / RQT / Rviz / Foxglove
- Docker / Docker Compose
- Nvidia Jetson /Arduino / Raspberry Pi / MiTAC
- OpenCV / OpenPCL / Apriltag / YOLO
- OpenVino / Pytorch / Optuna / LabelStudio
- WSL / Ubuntu / Yocto
- Github / Gitlab
- VS Code / Gimp
- Solidworks / Matlab

Interests

Japanese language

I've been learning by myself during my spare time.
My current goal is to pass the JLPT N3 exam.

Sports

I love hiking, cycling, kayaking... Any sort of group activity in natural landscapes. Also, I started taking Taekwondo classes last year.

Novels

I enjoy solving Agatha Christie's murder mysteries. Finally, I recently picked up on the books from "The Way of Kings" by Brian Sanderson.

Projects

From June 2025 up until now

Deployment of an online ROS laboratory
Personal project | Arlon, Belgium
I used the roslib javascript library to connect a web application to a Gazebo simulation running on my private server. The website is deployed on Cloudflare and the rosbridge websocket connection is established with Cloudflare tunneling.
- Real-time remote control and video feedback of the simulated turtlebot.
- Apriltag detection and YOLO segmentation of other robots.

December 2021

3D drawing tool with Virtual Reality controllers
Personal project | Thionville, France
I used an open-source medical robotics library to retrieve the pose of my Oculus Quest 2 controllers and draw their position in a 3D Python plot.

October 2021 to February 2022

Preliminary design of the NOIRE nanosatellite swarm
LESIA Laboratory | Meudon, France
I contributed to the specifications of the deployment conditions of a cubesat swarm used for interferometric measurements of FIR signals. I simulated each orbital path with Python to represent the evolution of the swarm configuration.

November 2020 to January 2021

Unicycle robot SLAM and path-finding algorithm
Polytech Sorbonne | Paris, France
I designed the motion control software of a mobile robot which I tested in Gazebo. With the help of SLAM ROS libraries and RRT path-finding algorithms, I managed to control the platform navigating in a room in fully autonomous mode.

October 2020 to January 2021

Implementation of a mobile robot with embedded sensors
Polytech Sorbonne | Paris, France
I programmed the drivers and navigation algorithm of a mobile robot using a Raspberry pi, hall effect sensors and odometric principles. I mounted a serial arm on it with a camera, using a Haar-Cascade-powered AI to track human faces.

September 2019 to June 2020

Design of a pan-tilt robotic system
Générations Robots | Paris, France
I worked as part of a team to design an IP67 motorized system with two rotation axes, capable of accommodating sensors for our client who wanted to integrate it into a mobile robot. I was especially working on the electrical blueprint.

January 2019 to May 2019

Prototyping of a robotic arm
Polytech Sorbonne | Paris, France
Within a team of Polytech students, I built a 2DOF parallel robotic arm from the CAD on Solidworks to the control algorithm on an Arduino. I used an IR sensor to plan the motion of its effector along the curves of an irregular wall.

Education

From 2022 to 2023

Linguistic and traineeship program "Vulcanus in Japan"
EU-Japan Centre for Industrial Cooperation | Tokyo, Japan
- 4 months of intensive Japanese courses at the Naganuma School with fellow European participants followed by an 8-month internship in a Japanese company.

From 2021 to 2022

Master's degree "Astronomical and Space-based Systems Engineering"
Paris Observatory, Sorbonne University | Paris, France
- Space-based embedded systems engineering, sensors, optics, machine learning (pattern recognition, optimization...), astrophysics, orbitography, radiation.

From 2018 to 2021

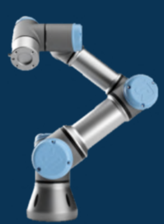
Robotics Engineering Degree
Polytech Sorbonne Engineering School | Paris, France
- *3rd year:* mobile robotics (ROS, SLAM, Kalman filter, pathfinding...), real-time and object-oriented programming, test-driven development, control of high-inertia robotic arms, AI, machine learning (deep neural network, SVM...).
- *2nd year:* ethics, analogical and numerical automatics, robotics modelization (Denavit-Hartenberg, forward and reverse kinematics, state-space representation, trajectory planification...), image and signal processing and computer vision.
- *1st year:* control and power electronics, mechanical conception, mathematics.

Robot collection

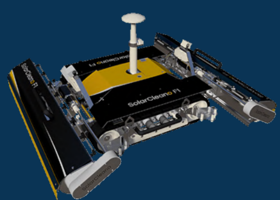

P-Guard


S-Drone


S-Buoy


Ur5e

B1A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